

Intro to Programming Logic

T.U.B.U.R.A.N. Learning Module

Variables

A variable is a named container that stores a piece of data, such as a number, text, or true/false value. Variables let a program remember and reuse information, like a sensor reading or a counter.

Conditionals (If/Else)

Conditional statements let a program make decisions. An 'if' statement checks whether something is true, and runs a block of code only if it is. An 'else' block runs instead when the condition is false. This is how a robot 'decides' what to do.

Loops

A loop repeats a set of instructions multiple times, without needing to write the same code over and over. Common loop types include 'for' loops (repeat a fixed number of times) and 'while' loops (repeat as long as a condition remains true).

Putting Logic Together

A simple robot behavior might combine all three: a variable stores a sensor reading, a conditional checks if the reading crosses a threshold, and a loop keeps checking this continuously so the robot can react in real time.

Why This Matters

These three concepts, variables, conditionals, and loops, are the foundation of nearly all programming, whether you are coding a simple robot, a mobile app, or a website. Mastering them early makes every future programming topic easier to learn.